

A Compositional Calculus for Cross-Domain Scientific Authority

Sze Chun Yiu
Stockholm University
sze-chun.yiu@fysik.su.se

31 August 2026

Abstract

Local authorization, delegation, verification, and claim-evidence mechanisms do not by themselves establish a different scientific obligation. We introduce a bounded scientific-authority lifting and composition calculus spanning thirteen donor families. Target obligations are typed by domain, kind, scope, content, and epoch. Donor-native authority is conserved, and scientific authority propagates only through type preservation or narrowing, or through an explicit subject- and epoch-bound protected coercion. Blockers remain three-valued: REFUTED, UNDETERMINED, and ESTABLISHED; UNDETERMINED yields CANNOT_CHECK. Alternative complete support families make revocation exact. A finite model contains 3,072 authority states and 39,936 donor-family evaluations; a second implementation reproduces the canonical enumeration. The thirteen families are a replication factor, not a state dimension, and the result is equivalent to an ideal information-matched product. The contribution is a bounded cross-domain authority-composition semantics. It does not claim generic authorization, ownership of donor mechanisms, local scientific verification, deployed-agent performance, centralized expressive superiority, or naturalistic evidence of effectiveness.

Overview and result

Autonomous scientific systems increasingly operate under strong local authority mechanisms. Agent-governance work provides evidence-backed permission graphs, proof-carrying action certificates, cryptographic delegation provenance, deterministic authority narrowing, accumulated principal-chain authorization, typed verifier certificates and heterogeneous authorization-evidence chains. Scientific-agent systems separately provide evidence-bound release gates, claim-to-evidence verification, contract-governed research artifacts, evidence-calibrated claim adjudication and evidence-ledger review. This paper treats all of these mechanisms as donor-owned local authorities rather than claiming generic authorization or scientific verification as new.

The remaining problem is compositional: when may a local authorization, delegation, verifier, claim-evidence or release verdict establish a different target scientific obligation? This paper introduces a bounded **scientific-authority lifting and composition calculus** over thirteen donor families. The target obligation is typed by (domain, kind, scope, content, epoch). Native donor authority is conserved. Scientific authority propagates across heterogeneous chains only when the scientific type is preserved or narrowed, or when an explicit subject/epoch-bound protected coercion supplies a complete bridge. Blockers use three states: REFUTED, UNDETERMINED, and ESTABLISHED, with UNDETERMINED yielding CANNOT_CHECK rather than silently disappearing. Alternative complete support families make revocation exact: invalidating one route preserves authority when another complete independent route survives, while destroying all complete routes removes discharge. Action or release permission and scientific support are explicitly non-inverse relations.

The final X4 finite model covers 3,072 distinct exact authority states, each replayed across thirteen donor families for a total of 39,936 evaluations. The distinction matters and we state it plainly: the terminal function takes seven arguments and the donor family is not among them, so over the 239,616 pairs of evaluations differing only in donor, none changes the verdict. That invariance is the donor-conservativity result rather than additional coverage, and the thirteen families are a replication factor, not a state dimension. It has zero donor-conservativity violations and zero mismatches with an ideal equally typed decentralized product. It contains 65 minimal scientific-type separation witnesses, 65 protected-coercion successes and 65 matched unprotected countermodels, the three-state blocker law for all thirteen donor families, 26 single-support-family revocation survivals and 13 all-support-family blocks, plus 169 heterogeneous ordered-chain composition successes and 169 matched scientific-authority-widening countermodels. A second implementation independently reproduces the canonical enumeration. The result is a bounded cross-domain scientific-authority composition semantics, not generic authorization, local scientific verification, deployed-agent superiority, or a claim of centralized expressive advantage.

Donor-engulfment architecture

V3 explicitly imports the strongest useful structure from two neighboring layers.

Action, delegation, and authorization donors

P8 reuses evidence-backed permission graphs and external deterministic authorizers; runtime-neutral action certificates carrying action identity, approvals, runtime and outcome receipts; append-only human-to-agent delegation provenance; authority-narrowing and cascade-containment rules; principal-chain composition with bounded scope; typed verifier certificates; heterogeneous authorization-evidence chains; and generic cross-domain authority relations.

These mechanisms answer important local questions such as whether an action is permitted, whether a delegation is valid, whether required evidence certificates are present, or whether heterogeneous authorization receipts jointly satisfy an action policy. This paper claims none of those mechanisms as its own.

Scientific claim, release, and adjudication donors

P8 also reuses domain scientific harnesses that separate semantic from scientific authority and authorize evidence-bound release; claim-to-evidence chains with deterministic grounding and verification; persistent contract-governed research artifacts; evidence-calibrated claim adjudication; and evidence-ledger review that routes unsupported, contradicted, or mixed-evidence claims back for revision.

These donors establish that scientific claim validation/release can itself be externalized and governed. P8 therefore does not claim novelty merely for separating a model’s fluent answer from a scientific release decision.

The remaining composition problem

The donor mechanisms are powerful but heterogeneous. Their local verdicts concern different objects: an action, a delegation, a receipt chain, a verifier predicate, a claim-evidence packet, a research artifact, or a domain-specific release decision. This paper’s V3 object is the common

relation that governs when such a local authority object is entitled to discharge a target scientific obligation of another type or scope.

Scientific-discharge type and lifted terminal

The complete scientific-discharge type is

`tau = (domain, kind, scope, content, epoch).`

Let $D(a)$ denote the native authority, verification, adjudication, or release verdict of donor object a . This paper does not rewrite D .

The bounded lifted state additionally contains:

- **narrowing_ok**: every relevant authority/delegation/claim-processing hop preserves or narrows `tau`;
- blocker state `b` in `{REFUTED, UNDETERMINED, ESTABLISHED}`;
- two alternative complete support families `S_A`, `S_B`;
- optional explicit **protected_coercion** binding a source type to the target type and epoch.

A donor-derived judgment reaches **DISCHARGE** only when:

1. $D(a)$ is valid;
2. the scientific type matches directly or a complete protected coercion is registered;
3. authority is non-widening across the relevant chain;
4. blocker state is **REFUTED**;
5. at least one complete support family survives.

UNDETERMINED maps to **CANNOT_CHECK**; **ESTABLISHED** maps to **BLOCK**.

Theorem V3.1 — donor conservativity

Adding the scientific-discharge layer does not change any donor-native action, delegation, evidence, adjudication, or release verdict. Local authorities remain usable for their native purpose even when they do not discharge a broader scientific target.

Theorem V3.2 — cross-donor scientific-discharge separation

For every registered donor family and each non-inert coordinate of `(domain, kind, scope, content, epoch)`, native donor success can coexist with scientific-discharge failure when only that coordinate is incompatible with the target obligation.

Thus a valid action certificate, verified claim-evidence packet, scientific release decision, or cross-domain authorization may be retained without being treated as an unrestricted authority token.

Theorem V3.3 — monotone scientific authority propagation

Scientific authority may be preserved or narrowed through heterogeneous authority chains. A hop that widens the scientific type beyond inherited authority cannot discharge the widened target merely because each local donor step remains native-valid.

This extends donor narrowing/cascade-containment ideas to the scientific-obligation type rather than replacing them.

Theorem V3.4 — protected coercion

A complete protected coercion may transform an otherwise incompatible scientific type. The same mismatch without the registered bridge remains blocked. Semantic similarity, intent similarity, common action digest, claim fluency, or ordinary action composability is not sufficient to create a coercion.

Theorem V3.5 — three-state blockers

Under otherwise satisfied conditions:

- `REFUTED` permits the blocker condition to clear and allows discharge;
- `UNDETERMINED` yields `CANNOT_CHECK`;
- `ESTABLISHED` blocks discharge.

The distinction prevents unknown blocker state from being silently treated as blocker absence.

Theorem V3.6 — exact support-family revocation

When two independent complete support families establish the same target obligation, revoking one family preserves discharge through the other. Revoking all complete support families removes discharge.

This avoids both over-revocation, in which one failed premise destroys an independently supported result, and under-revocation, in which a broken unique support path is ignored.

Theorem V3.7 — heterogeneous authority-chain composition

Every ordered pair among the thirteen registered donor families admits a scientifically valid two-hop composition under native-valid local verdicts, compatible or protectedly bridged scientific type, non-widening authority, refuted blockers, and surviving support. The matched chain with a scientific-authority-widening hop is blocked even though both donor-native steps remain valid.

The theorem treats heterogeneous authority composition as a first-class object rather than requiring every donor to adopt one common receipt format.

Theorem V3.8 — permission, release, and support are not inverse relations

A native denial of one action or release route is not scientific refutation when an independent complete evidence family establishes the proposition. Conversely, native authorization or release does not authorize a different or broader scientific target. The two relations are connected by explicit evidence/type bridges rather than logical inversion.

Theorem V3.9 — heterogeneous receipt composition is a prerequisite, not scientific sufficiency

Valid composition of authorization receipts, canonical action or subject binding, and successful local policy evaluation may be necessary inputs to a scientific claim, but they do not by themselves establish target scientific sufficiency. Scientific discharge remains governed by the target type, blocker, support, and coercion relation.

Theorem V3.10 — ideal decentralized-product equivalence

A decentralized donor product supplied with the exact same scientific type, narrowing, blocker, support-family, coercion, and composition rules agrees extensionally with P8. The contribution is therefore the explicit common scientific-discharge calculus, not centralized expressive power.

Final donor envelope and literature fixed point

The final donor envelope contains thirteen families across authorization/security and scientific verification/adjudication. The research chronology is retained deliberately: early six-donor and ten-donor intermediate enumerations are preserved, because later hostile searches discovered stronger scientific-release and heterogeneous-authority donors and forced the architecture to expand rather than shrink.

Two post-X4 search rounds produced no further material change to the interface. This is a bounded current-pass fixed point, not a claim of global or exhaustive literature saturation. A newly discovered primary formalism with an equivalent target scientific-discharge relation is a reopen trigger.

Exact bounded support

Final X4 enumeration:

- distinct exact authority states: **3,072**, each replayed across thirteen donor families for **39,936** evaluations; the donor axis changes no verdict in any of 239,616 sibling pairs, which is the conservativity result and not extra coverage;
- terminals: **19,968** NO_DONOR_AUTHORITY, **15,353** BLOCK, **3,328** CANNOT_CHECK, **1,287** DISCHARGE;
- donor-conservativity violations: **0**;

- minimal scientific-type separation witnesses: **65**;
- protected-coercion successes: **65**;
- matched unprotected-coercion countermodels: **65**;
- blocker-law instances: **13** REFUTED successes, **13** UNDETERMINED CANNOT_CHECK, **13** ESTABLISHED blocks;
- one-support-family revocation survivals: **26**;
- all-support-family revocation blocks: **13**;
- heterogeneous ordered-chain composition successes: **169** — one composition counted thirteen times thirteen, because the enumeration’s chain loop ignores both of its donor variables and the thirteen families present a single profile to it; the mechanized derivation below states what that leaves;
- matched scientific-authority-widening chain countermodels: **169**, on the same replication;
- action/release-denied but independently supportable proposition examples: **13**;
- ideal decentralized-product mismatches: **0**;
- canonical row SHA-256:
ed186b824692fd5b3ab31be718c75b84e2126b577ce921ca5cc01b2d08ae19e6.

A separate checker independently reconstructs the final enumeration.

Mechanized core, and what the finite result is an instance of

Everything above this line is an enumeration over an authored state space. That is a true statement about 3,072 states and thirteen donor families, and it is not a theorem. Three machine-checked artifacts under `formal/mechanized/` now say which parts of it follow from a general semantics, and what each part costs.

The calculus. Ten composition and revocation theorems — scope non-amplification, domain confinement, non-laundering, non-compensatory obligations, defeater monotonicity, epoch isolation, the one-step delegation lemma, a bridge from conversion to reachability, and a characterisation of cycles — are discharged by Z3 over uninterpreted domain, object and issuer sorts, so none of them is a statement about thirteen donors or six named domains. Two limits travel with them and are recorded in the artifact rather than in a footnote: reachability is *axiomatised* as a reflexive-transitive closure carrying a well-founded rank, because transitive closure is not first-order definable, and the chain theorem’s induction schema is the single hand step in the development. The rank is not a formality — an earlier axiomatisation without it admitted circular reachability, in which a sink domain reached its own source, and a differential against the committed checker is what caught it.

The state model as an instance. Each of the seven arguments the X4 terminal takes is assigned to the calculus conjunct it means: type-coordinate agreement or a registered protected coercion is the calculus’s reachability, narrowing is scope containment, and the blocker is at once a hard obligation and a defeater. The calculus’s own rule under that assignment reproduces X4 exactly on all 3,072 distinct states, with all four terminals reached.

The chain compositions as instances. A donor family is interpreted as an element of an uninterpreted sort whose *domain is its scientific type profile*; a protected coercion is a registered conversion and narrowing is scope containment. Under eight frame conditions stated as axioms — each shown load-bearing by exhibiting a countermodel to a theorem it carries — seven theorems are discharged, and donor-level chain confinement is expanded and discharged at every chain length up to six. The bound is a measured limit rather than a chosen one: lengths seven to twelve were tried and the solver returned **unknown** on some runs and a proof in seconds on others, so the ladder stops where it is a reliable corroboration. Nothing about the theorem changes at length seven — the induction schema is what carries every length, and it is the single hand step in the development. A P8 chain hop *is* the calculus’s delegation, which is itself one of the discharged theorems, so Theorem V3.7 is an instance of a statement that mentions neither thirteen nor two.

What that costs Theorem V3.7. The derivation lands, and what it lands on is small. We state it here rather than leave a reader to assume otherwise.

- The 169 is one composition counted 169 times. The committed enumeration writes its chain claim as a doubly nested loop over the thirteen families whose body mentions neither loop variable, and under the interpretation all 169 ordered pairs compose to a single distinct state. The donor axis is a replication factor here exactly as it is in the 39,936 evaluations, and here it is squared.
- Nothing heterogeneous is exercised. The state the chain claim evaluates has all five type coordinates agreeing, which by the interpretation puts both donors in one domain; the hop is then a reach by reflexivity and no conversion is ever consulted. Read the thirteen families as type-distinct, so that a cross-family hop crosses a type boundary unless a protected bridge is registered, and the same committed rule returns **13** compositions rather than 169. Registering every cross-family bridge returns 169 again. The published number therefore records the reading under which no two of the thirteen donor families differ in scientific type — which is the reading under which the chain is not heterogeneous.
- Neither published count tests the interpretation. All eight wrong composition operators tried reproduce 169 successes exactly, and six of the eight also reproduce the 169 widening countermodels. What separates them is an exhaustive identity checked through the committed rule on 36,864 representative pairs, which stand in exactly for all 9,437,184 pairs of X4 states: a composed chain discharges when and only when both of its hops do. Every one of the eight wrong operators breaks it, and the shipped rule satisfies it with no exception.

None of the above has been checked by anyone outside the lane that wrote it. Independent formal review and independent systems reproduction remain open, and no empirical claim is made or supported here.

Wider P8 claim

P8 composes modern action authorization, delegation provenance, heterogeneous authorization receipts, cross-domain authority relations, typed verifier certificates, scientific release gates, claim-evidence chains, research harnesses, claim adjudication and evidence-ledger review into a common scientific-discharge calculus. Local donor authority remains intact, while authority to establish a scientific conclusion propagates only through typed non-widening chains or protected coercions, fail-closed blocker semantics and surviving complete support families.

This is deliberately wider than the V2.1 slogan that generic permission differs from scientific permission. The local authority mechanisms are absorbed; the paper’s contribution is the cross-domain composition/lifting relation among them.

Limits

V3 does not claim novelty for any of the thirteen donor mechanisms, universal minimality of the five type coordinates, deployed-agent superiority, a public full-manuscript release gate, or global literature saturation. It does not claim that the 169 chain compositions measure heterogeneity, that they are 169 results rather than one replicated, or that reproducing them validates any interpretation of the calculus; the mechanized section above states what each of those numbers was found to carry. Native authorization failure is not scientific refutation. The ideal equally typed decentralized product ties exactly.

Replacement conclusion for V3

This paper’s strongest form is a composition theory rather than another authorization gate. Modern agent systems already provide serious local authority: permission graphs, proof-carrying actions, delegation chains, typed verifier certificates, receipt composition, domain scientific release, claim-evidence verification, research harnesses, and evidence-ledger adjudication. ORION should use those mechanisms rather than rebuild weaker versions of them.

The missing interface is the authority relation between these local judgments and a target scientific obligation that may differ in domain, kind, scope, content, or epoch. P8 makes that relation explicit. Scientific authority is non-widening by default, transformable only by protected coercion, fail-closed under unresolved blockers, exactly revocable across alternative complete support families, and compositionally reusable across heterogeneous donor systems. An equally typed decentralized implementation has the same expressive power, which keeps the contribution architectural and semantic rather than brand-dependent.

Current science terminal:

P8_CROSS_DOMAIN_SCIENTIFIC_AUTHORITY_COMPOSITION_SUPPORTED__13_DONOR_FORMAL_ENVELOPE__IDEAL_PRODUCT_EQUIVALENT.

References

- [1] Action evidence boundary for consequential agent effects, 2026. IETF Internet-Draft.
- [2] A. Alvarez, S. Rajan, S. Mugel, and R. Orús. ProvenanceGuard: source-aware factuality verification for MCP-based LLM agents, 2026. arXiv:2606.18037.
- [3] Authorization evidence chains, 2026. IETF Internet-Draft.
- [4] Moritz Y. Becker, Cédric Fournet, and Andrew D. Gordon. SecPAL: design and semantics of a decentralized authorization language. *Journal of Computer Security*, 18(4):619–665, 2010.
- [5] A. Kaul, Q. Lan, and P. Gupta. AgentBound: verifiable behavioral governance for autonomous AI agents, 2026. arXiv:2606.30970.
- [6] X. Liu et al. AgentAbstain: Do LLM agents know when not to act?, 2026. arXiv:2607.10059.

- [7] Jaehong Park and Ravi Sandhu. Towards usage control models: beyond traditional access control. In *SACMAT*, pages 57–64, 2002.
- [8] Jaehong Park and Ravi Sandhu. The UCONABC usage control model. *ACM Transactions on Information and System Security*, 7(1):128–174, 2004.
- [9] O. Serdar and C. Mertayak. SteerBench-Work: A benchmark for agent steering at action boundaries, 2026. arXiv:2608.12654.
- [10] K. Tallam. Authorization propagation in multi-agent AI systems: identity governance as infrastructure, 2026. arXiv:2605.05440.
- [11] H. Tan, Y. Wang, P. Zhang, S. Li, and J. Shen. ETAS: An effect-typed language for agent systems, 2026. arXiv:2607.17780.
- [12] R. Tang, Q. Liu, Y. Zhang, Y. Wang, X. Chen, and C. Dong. Context is not authority: Structured runtime governance for financial market agents, 2026. arXiv:2608.09025.
- [13] Y. Zhang, X. Zhao, S. Liu, H. Lou, G. Cheng, and C. Liu. FAVA: Formal authorization for verified agents with evidence-backed permission graphs, 2026. arXiv:2607.27267.